

Title: \_\_\_\_\_

Name: \_\_\_\_\_

Date: \_\_\_\_\_

## IS - Intelligent System

→ Related AI subject portion

→ Language Use :- PROLOG

Lab - 1 (19<sup>th</sup> Jan, 2026) , Introduction to Prolog

To do :- Implement DFS and BFS  
in any language.

Prolog :-

is a logic programming lang.  
used in AI.

It consist of rules and facts

These define  
relationship b/w facts

Title: \_\_\_\_\_

Name: \_\_\_\_\_

Date: \_\_\_\_\_

eg:-

Grandfather (X, Y) :- father (X, Z), parent (Z, Y)



Queries:- Questions that we ask Prolog

ex:- ?likes(john, mango)

↳ is a query asks  
if john likes mango.

Title: \_\_\_\_\_

Name: \_\_\_\_\_

Date: \_\_\_\_\_

→ Hair is black

black (Hair)

→ Nawaz loves to play games

lovestoplay (Nawaz, games)

→ Tom is a cat

Cat (Tom)

---

→ If  $(:-)$   $(;)$  or

Suppose a clause is like:

$P :- Q; R.$

$\hookrightarrow P :- Q.$

$P :- R.$

Title: \_\_\_\_\_

Name: \_\_\_\_\_

Date: \_\_\_\_\_

eg:- Lili is happy if she dances.

↓  
:-

happy(lili) :- dances(lili).

2. Tom is hungry if he is searching for food

hungry(Tom) :- searching-for-food(Tom).

3. Jack and Bili are friends if both of them love  
to play cricket

friends(jack, bili) :- love-to-play-cricket(jack, bili).

or

love(cricket(jack), lovesCricket(bili)